Cambridge Assessment International Education

Cambridge IGCSE™

BIOLOGY **0610/13**

Paper 1 Multiple Choice (Core) **May/June 2021**

45 minutes

You must answer on the multiple choice answer sheet.

You will need: Multiple choice answer sheet
Soft clean eraser
Soft pencil (type B or HB is recommended)

INSTRUCTIONS

- There are **forty** questions on this paper. Answer **all** questions.
- For each question there are four possible answers **A**, **B**, **C** and **D**. Choose the **one** you consider correct and record your choice in soft pencil on the multiple choice answer sheet.
- Follow the instructions on the multiple choice answer sheet.
- Write in soft pencil.
- Write your name, centre number and candidate number on the multiple choice answer sheet in the spaces provided unless this has been done for you.
- Do **not** use correction fluid.
- Do **not** write on any bar codes.
- You may use a calculator.

INFORMATION

- The total mark for this paper is 40.
- Each correct answer will score one mark.
- Any rough working should be done on this question paper.

This document has **16** pages. Any blank pages are indicated.

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1 What is a characteristic of all living organisms?

A breathing

B circulation

C egestion

D sensitivity

2 Which statement about the binomial naming system is correct?

A It is used for naming animal species but not plant species.

B It is an internationally agreed scientific naming system.

C The name of a species is different in different countries.

D The name consists of one or two words.

3 Scientists discover a new species of animal.

It has a segmented body with two pairs of legs on each segment.

To which group of animals does this new species belong?

A arachnids

B crustaceans

C insects

D myriapods

4 In which part of a cell does photosynthesis take place?

A vacuole

B nucleus

C chloroplast

D cytoplasm

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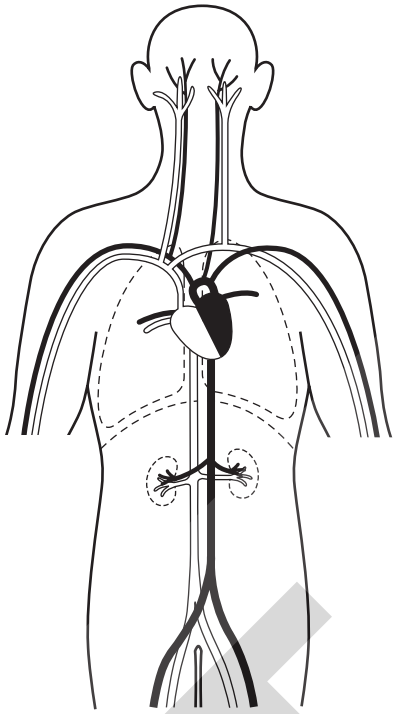
5 The diagram shows some of the blood vessels and other structures in the human body.



The blood vessels shown are all parts of the same

A cell.

B organ.

C organ system.

D tissue.

6 A photograph shows a plant cell nucleus measuring 2 mm across.

If the magnification of the cell is ×500, what is the actual size of the nucleus?

A 0.00002 mm **B** 0.004 mm **C** 0.04 mm **D** 250 mm

7 By which process do oxygen and carbon dioxide move between cells and capillaries?

A breathing

B diffusion

C excretion

D respiration

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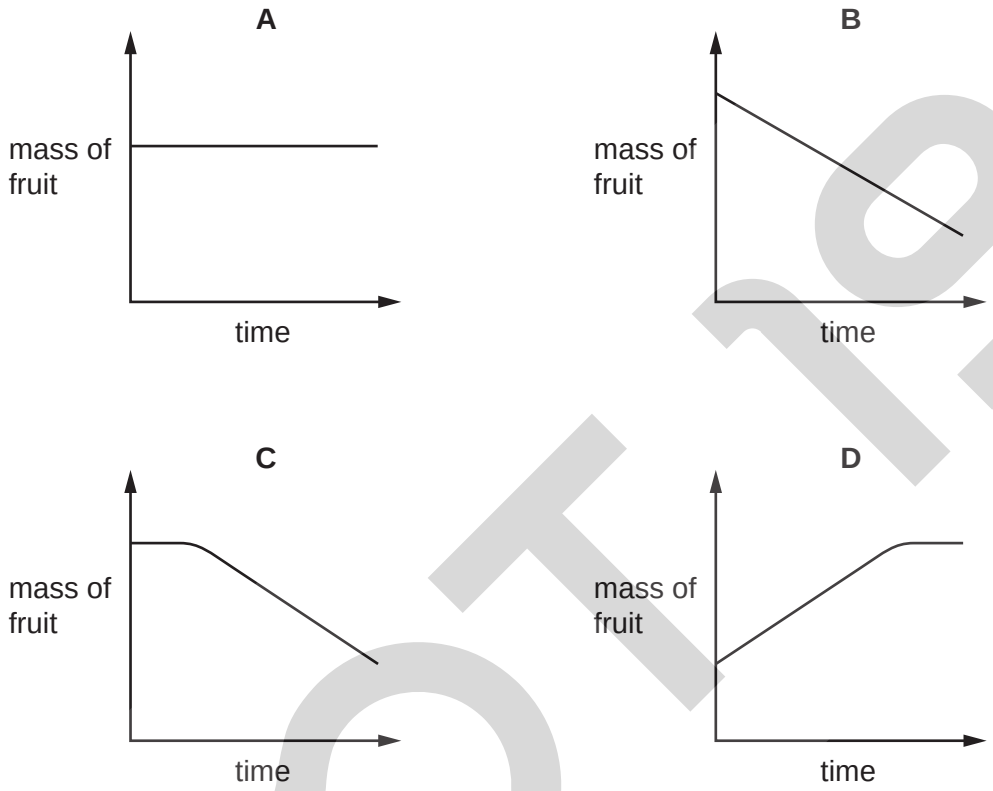
8 Some students investigated osmosis.

They placed some dried fruit into a beaker of distilled water. The mass of the fruit was measured at regular intervals during a 24-hour period.

Which graph shows the results?



9 Which element is found in proteins but **not** carbohydrates?

A carbon

B hydrogen

C nitrogen

D oxygen

10 Starch is digested by amylase in the mouth, but it is not digested in the stomach.

What is the reason for this?

A All starch digestion is completed in the mouth.

B The pH in the stomach is not suitable for the amylase to work.

C The starch does not stay in the stomach long enough to be digested.

D The temperature in the stomach is not suitable for the amylase to work.

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11 Which type of molecule are enzymes made of?

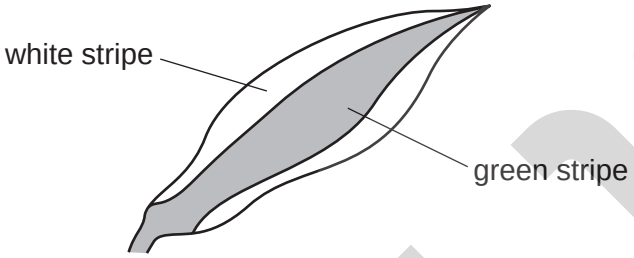
A carbohydrate

B fat

C protein

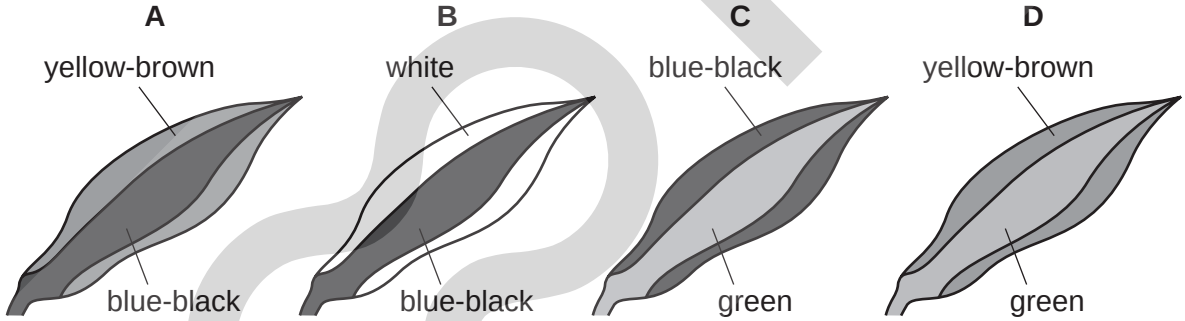
D vitamin

12 A plant with striped leaves was kept in bright light for six hours.



A leaf was taken from the plant and the chlorophyll was removed. The leaf was then tested for starch using iodine solution.

Which diagram shows the result of the test?



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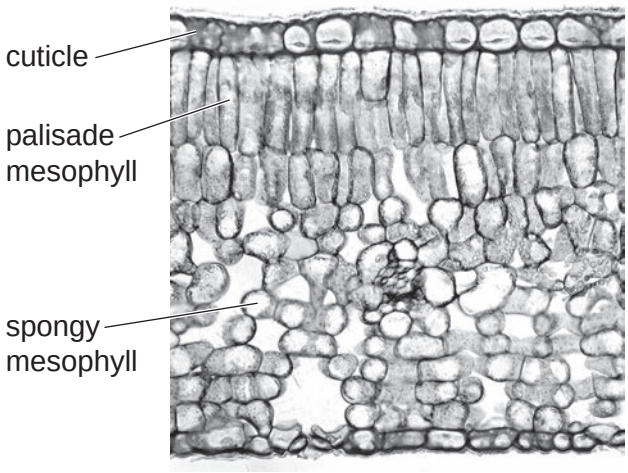
13 Which photomicrograph of the leaf section has been correctly labelled?

A



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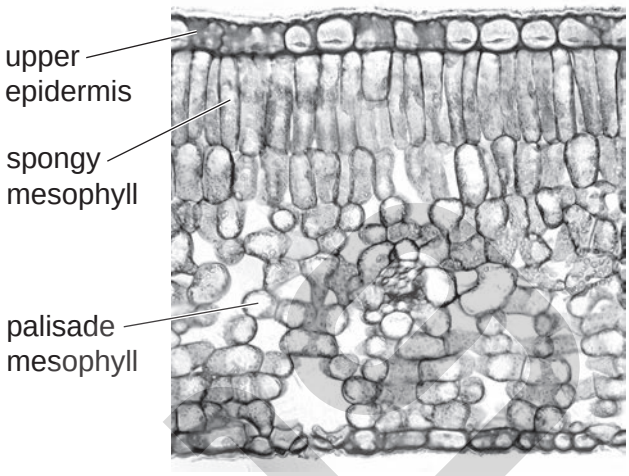

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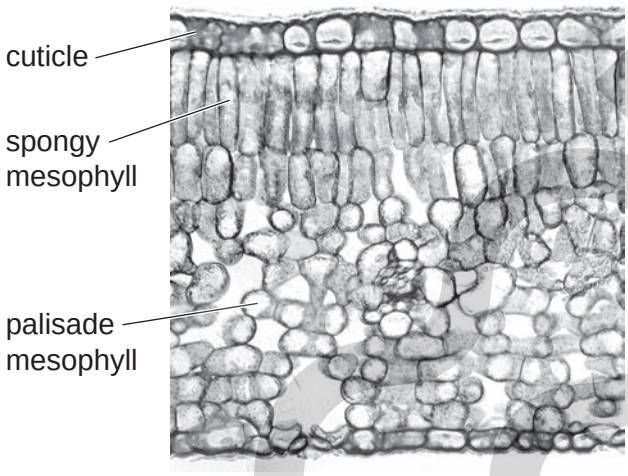

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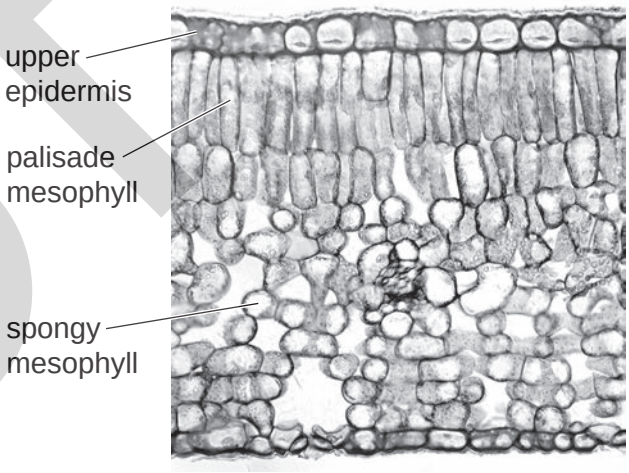

14 In humans, where does most of the absorption of digested food take place?

A colon

B kidney

C liver

D small intestine

15 What is the cause of the **start** of dental decay?

A Acid dissolves dentine.

B Acid dissolves enamel.

C Sugars dissolve dentine.

D Sugars dissolve enamel.

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16 Which statement correctly identifies the enzyme and the products of fat digestion?

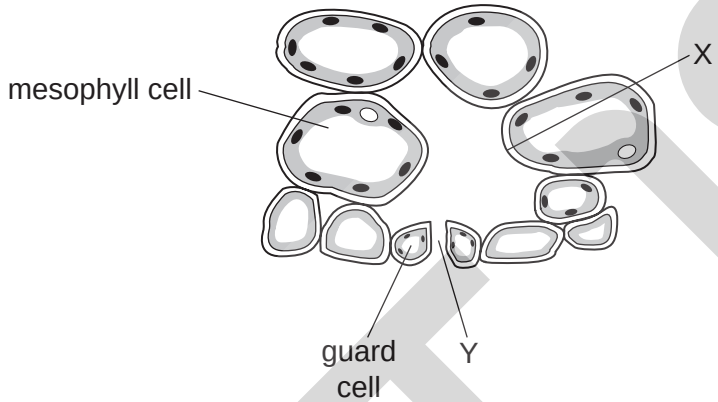
A Fats are digested by lipase to form amino acids and glycerol.

B Fats are digested by lipase to form fatty acids and glycerol.

C Fats are digested by protease to form amino acids.

D Fats are digested by protease to form fatty acids and glycerol.

17 The diagram shows one of the stomata of a leaf, and some of the cells that are near it.



During transpiration, what describes the movement of water at X and at Y?

	movement of water at X	movement of water at Y
A	evaporation	diffusion
B	evaporation	osmosis
C	osmosis	diffusion
D	osmosis	osmosis

18 A plant shoot is placed in a solution of dye.

The dye moves up the stem.

Under which conditions will the dye move most slowly?

	temperature	humidity
A	high	high
B	high	low
C	low	high
D	low	low

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19 Through which blood vessels does blood flow into the heart?

1 pulmonary artery

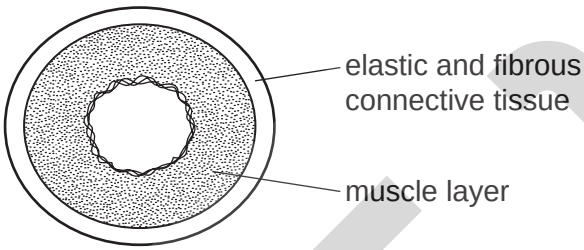
2 pulmonary vein

3 vena cava

4 aorta

A 1 and 2 **B** 1 and 4 **C** 2 and 3 **D** 2 and 4

20 The diagram shows a cross-section through a human blood vessel.



Which type of blood vessel does the diagram show?

A an artery

B a capillary

C a vein

D a ventricle

21 What is a function of some white blood cells?

A to carry glucose

B to carry oxygen

C to produce antibiotics

D to produce antibodies

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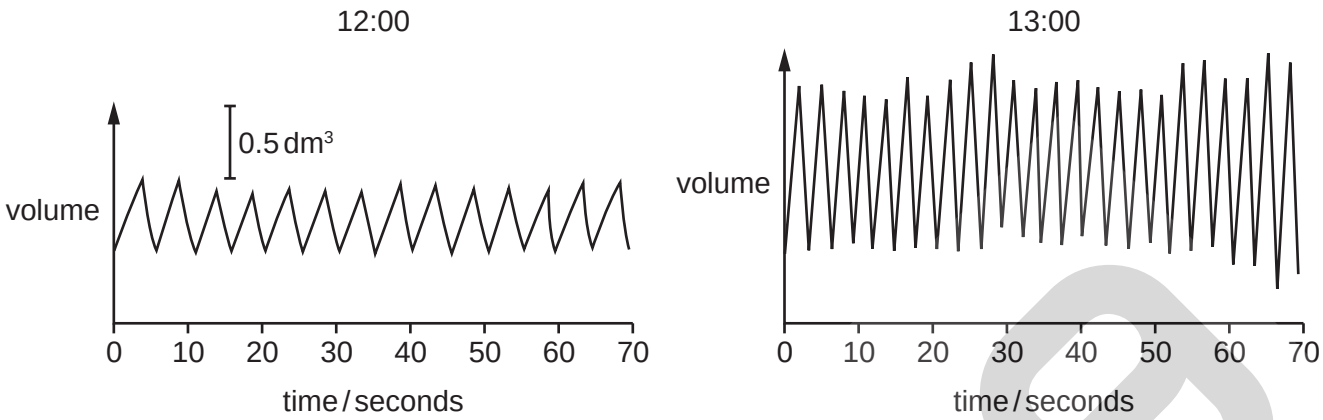
22 The diagrams show the depth and rate of breathing in a person at 12:00 and 13:00.



What happens to the person's breathing between 12:00 and 13:00?

	depth of breathing	rate of breathing
A	decreases	decreases
B	decreases	increases
C	increases	decreases
D	increases	increases

23 Which statement applies to respiration?

A It involves enzymes.

B Energy is absorbed.

C It only takes place in animal cells.

D Oxygen is released.

24 The table shows some products of respiration.

Which row shows the products of anaerobic respiration in muscles?

	alcohol	carbon dioxide	lactic acid	water
A	✓	✓	✗	✗
B	✗	✓	✗	✓
C	✗	✗	✓	✗
D	✗	✗	✓	✓

key
✓ = yes
✗ = no

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25 A student carried out an experiment to investigate the effect of temperature on the volume of urine produced.

Which row shows the experiment where the environmental temperature was increased from 20 °C to 40 °C but no other changes were made?

	urine produced / cm^3 per hour	
	before	after
A	60	60
B	80	40
C	120	145
D	100	130

26 A student's eyes blinked as a ball flew past their head. This is a reflex action.

Which row describes the reaction shown by the student?

A a fast reaction involving the endocrine system

B a fast reaction involving the nervous system

C a slow reaction involving the endocrine system

D a slow reaction involving the nervous system

27 What type of cell must an organ contain to detect a stimulus?

A blood

B receptor

C skin

D xylem

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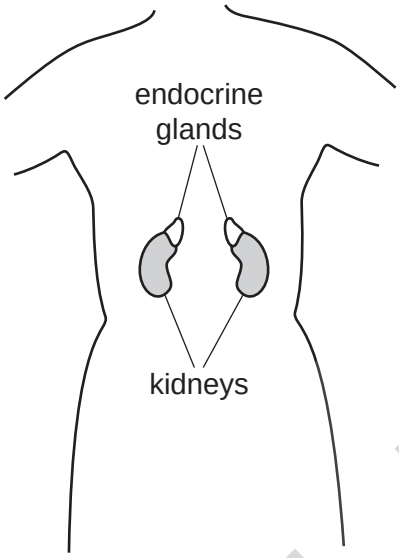
28 The diagram shows the position of the kidneys in a human. Each kidney has an endocrine gland above it.



Which glands are shown?

A adrenal glands

B ovaries

C pancreas

D testes

29 Why are people advised **not** to drive a car after drinking an excessive quantity of alcohol?

A Alcohol is a depressant.

B Alcohol is a stimulant.

C Alcohol decreases reaction time.

D Alcohol is addictive.

30 Which statement about the process of asexual reproduction is correct?

A It produces genetically different offspring.

B It requires two parents.

C It produces gametes.

D It requires only one parent.

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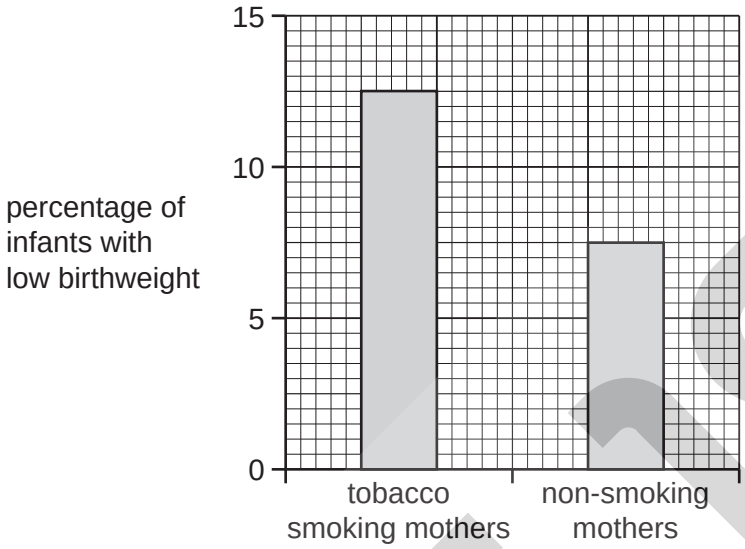
31 Scientists studied the effect on infant birthweight of mothers smoking tobacco. They compared their findings with the birthweights of infants from non-smoking mothers. The results are shown in the bar chart.



Which statement describes what this bar chart shows?

A 15% of infants born to mothers who smoke have a low birthweight.

B Mothers who smoke have 5% more low birthweight infants than non-smoking mothers.

C Non-smoking mothers have 10% fewer low birthweight infants than smoking mothers.

D Tobacco smoking is the only factor that has an effect on birthweight.

32 Changes occur in a boy's body at puberty; his height increases, more muscle develops and body hair grows.

Which hormone stimulates these changes?

A adrenaline

B insulin

C oestrogen

D testosterone

33 By which process is genetic information transmitted from generation to generation?

A inheritance

B mitosis

C meiosis

D variation

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34 Some reptiles can lose and regrow their tails to avoid predation.

Which process occurs to regrow the tail?

A digestion

B fertilisation

C meiosis

D mitosis

35 What is an adaptive feature of an organism?

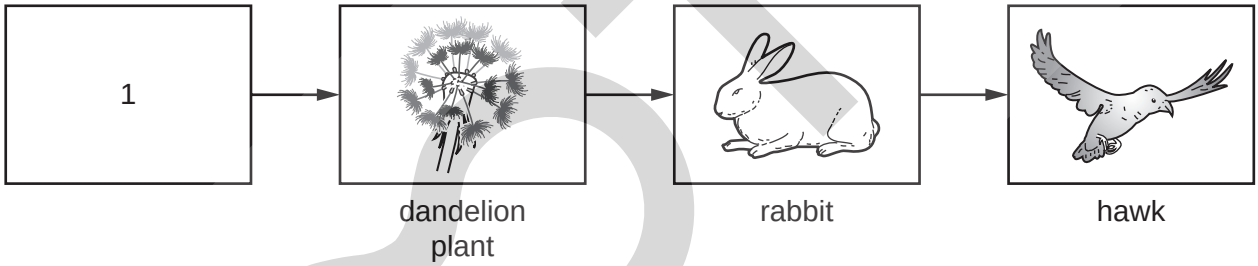
A any feature that is not changed by the environment

B any inherited feature that helps an organism to survive and reproduce

C any feature that shows discontinuous variation

D any inherited feature that an organism has

36 The diagram shows the energy flow through part of a biological system.



What should be in the box labelled 1?

A heat

B consumer

C producer

D the Sun

37 Human activities are changing the carbon dioxide concentration in the atmosphere.

Which activities will increase the carbon dioxide concentration in the atmosphere?

1 burning fossil fuels

2 burning cut-down trees

3 fossilisation of cut-down trees

4 planting replacement trees

A 1 and 2 **B** 1 and 3 **C** 2 and 4 **D** 3 and 4

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38 Which description of a population is correct?

A all the birds in a rainforest

B all the arachnids in a garden

C all the arthropods in the world

D all the red deer in a forest

39 What is an example of genetic engineering?

A the addition of vitamins to food to increase the nutritional benefits

B the breeding of sheep with thick wool by choosing animals with desired characteristics

C the insertion of genes into crop plants to make them resistant to insect pests

D the production of biofuels by yeast cells

40 What happens when deforestation takes place?

A The number of habitats is increased.

B The number of species becoming extinct is increased.

C The possibility of flooding is decreased.

D The volume of soil present is increased.

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